FAST Ultrasound Processing & 3D Visualization

Python Implementation of the FAST Framework Ultrasound Tutorial

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1. Project Overview

This project implements the concepts from the FAST Ultrasound Tutorial using portable Python libraries. The project demonstrates:

- Ultrasound data processing
- Image analysis
- 3D visualization
- Segmentation techniques
- Speckle tracking

Project Goal

Develop a Python-based implementation of FAST framework concepts for ultrasound image processing and advanced 3D visualization while maintaining portability across operating systems.

2. Project Modules

Module	Description
`01_read_and_visualize.py`	Reading and visualizing ultrasound data from various formats
`02_image_processing.py`	NLM denoising, scan conversion, envelope detection, and colormaps
`03_3d_visualization.py`	3D volume rendering, multi-planar reconstruction, iso-surface extraction, and 4D animation
`04_segmentation.py`	Tissue, vascular, and cardiac segmentation with contour detection
`05_speckle_tracking.py`	Block matching, displacement tracking, and strain estimation

3. Quick Start Guide

3.1 Python Environment Setup

```
cd src/ultrasound

# Create virtual environment
python3 -m venv venv
source venv/bin/activate

# Install dependencies
pip install -r requirements.txt
```

3.2 Run All Scripts

```
python 01_read_and_visualize.py
python 02_image_processing.py
python 03_3d_visualization.py
python 04_segmentation.py
python 05_speckle_tracking.py
```

3.3 Output

All generated images and animations are saved in the `output/` folder.

4. Running on Linux (Ubuntu)

4.1 Python Implementation

```
sudo apt-get update
sudo apt-get install -y python3-pip python3-venv \
libgl1-mesa-glx libxrender1

python3 -m venv venv
source venv/bin/activate

pip install -r requirements.txt
```

4.2 FAST Framework Direct Installation

```
pip install pyfast
```

Important Note

FAST framework supports Linux (Ubuntu 18.04+) and Windows. macOS is not officially supported.

5. Project Structure

```
src/ultrasound/
          01_read_and_visualize.py
          02_image_processing.py
          03_3d_visualization.py
          04_segmentation.py
          05_speckle_tracking.py
          requirements.txt
          utils/
                __init__.py
                data_generator.py
                image_utils.py
          output/
          report/
              REPORT.md
```

6. Dependencies

Package	Purpose
numpy	Array operations and computation
matplotlib	2D plotting and animation
scipy	Scientific computing and filtering
scikit-image	Image processing and marching cubes
pyvista	Interactive 3D visualization
vtk	3D rendering backend
Pillow	Image input and output
opencv-python	Image processing utilities

7. FAST Framework Mapping

FAST Component	Python Implementation
MovieStreamer / ImageFileStreamer	Synthetic data generation using `data_generator.py`
ImageFileImporter	NumPy array loading
NonLocalMeans	`nlm_fast()` implementation
ScanConverter	`scan_convert_sector()`
EnvelopeAndLogCompressor	`envelope_and_log_compress()`
Colormap.Ultrasound()	Custom ultrasound colormap generation
UltrasoundImageCropper	Ultrasound image cropping utilities
BlockMatching	Block matching implementation
AlphaBlendingVolumeRenderer	PyVista volume rendering
SlicerWindow	Multi-planar reconstruction
SegmentationNetwork	Classical segmentation techniques
display2D / SimpleWindow2D	Matplotlib-based visualization

8. Key Features

- Real-time ultrasound visualization
- Non-local means denoising
- Scan conversion and log compression
- Advanced 3D volume rendering
- Multi-planar reconstruction (MPR)
- Isosurface extraction
- Segmentation and contour detection
- Speckle tracking and strain estimation
- Cross-platform Python implementation

9. Conclusion

This project successfully demonstrates the implementation of FAST ultrasound processing concepts using portable Python libraries. The framework provides efficient visualization, segmentation, and tracking capabilities while maintaining compatibility across multiple operating systems.

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Report